

TQEC 1/28

Topologiq: J now upgrading the UI/UX, Severyn upgrading the algorithm

Action plan: Proposal: AugmentedNXGraph to encapsulate ZXGraph and BlockGraph

Make it easier to see what the algorithm does

Query graph, provide options based on properties of the ZX graph

Need logging, rewrite BFS, export to TQEC format

Alternative notation to help visualize (Δ_X , Y, Z)

Positional ZX graph presence in TQEC

Test suite

Test default: run for both $k=1$ and $k=2$

Just run $k=1$ by default, run $-k$ for both k

Run all conventions but NOT all variants

First step: By default, some tests are marked as slow ($k=2$),

Default: tests marked as slow do not run

Sam: run profiling too

Alexander Frei: new compiler options for lattice surgery

Surface code patching with layer trees in ZX notation